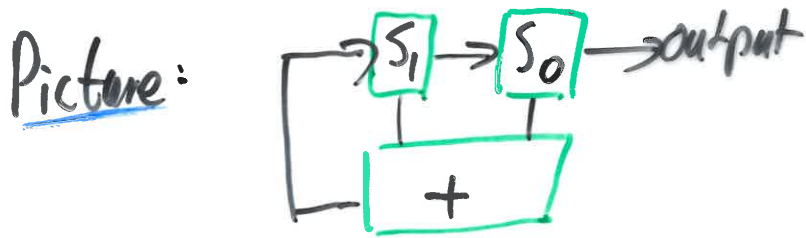


Ex: $L=2$, $f(a,b) = a+b$ (addition in $\mathbb{Z}_2$)

45



As a cryptosystem:

- Plaintexts = ciphertexts = bit streams
- Keys: initial states (s_1, s_0)
- Encryption: add corresponding bits of plaintext and key streams
- Decryption: $\xrightarrow{\text{plaintext}} \xrightarrow{\text{key stream}} \text{ciphertext}$

(Note: addition = subtraction over $\mathbb{Z}_2$)

For example:

	t	s_{t+1}	s_t
0		1	0
1		1	1
2		0	1
3		1	0
4		1	1
⋮		⋮	⋮

Time

state after t clock cycles

initial state = key

back to the initial state

In the example, the key stream (=right-most column) is 46

0 1 1 | 0 1 1 | 0 1 1 | ...

and encryption using the key $s_1=1, s_0=0$ e.g. operates as

plaintext	0	1	0	0	1	0	1	0	1	...
key stream	0	1	1	0	1	1	0	1	1	...
ciphertext	0	0	1	0	0	1	1	1	0	...

Note that our feedback shift register acts like a Vigenere cipher with key string 011 for the given key $s_1=1, s_0=0$.

Mathematical analysis of feedback shift registers

Given $L \geq 1$, a feedback function $f: \mathbb{Z}_2^L \rightarrow \mathbb{Z}_2$ and initial values $s_0, \dots, s_{L-1} \in \mathbb{Z}_2$, for $j \geq L$ we recursively define

$$s_j = f(s_{j-1}, \dots, s_{j-L}).$$

(Ex: $s_L = f(s_{L-1}, \dots, s_0)$, $s_{L+1} = f(s_L, \dots, s_1, \dots)$)

Then $s_0, s_1, \dots$ is the output stream of the feedback shift register of length L associated with f .

Goal: Find f (and $s_0, \dots, s_{L-1}$) such that the sequence $s_0, s_1, \dots$ "looks random".

Here's a serious obstacle:

Exercise: Given any L, f , and $s_0, \dots, s_{L-1}$ as above, the resulting sequence $s_0, s_1, \dots$ is always ultimately periodic in the following sense:

$$\exists c, N \geq 1. \forall j \geq c. s_j = s_{j+N}$$

We call the least possible value of such a number N the (least) period of the sequence.

Sketch of proof: Define a function $\text{adv}: \mathbb{Z}_2^L \rightarrow \mathbb{Z}_2^L$ ("advance state") via

$$\text{adv}(s_{L-1}, \dots, s_0) = (f(s_{L-1}, \dots, s_0), s_{L-1}, \dots, s_1).$$

Given an initial state $\sigma_0 = (s_{L-1}, \dots, s_0) \in \mathbb{Z}_2^L$, recursively define states $\sigma_0, \sigma_1, \dots$ via $\sigma_{j+1} := \text{adv}(\sigma_j)$.

Easy: the last (= rightmost) component of σ_j is s_j .

Hence: it's enough to show that $\sigma_0, \sigma_1, \dots$ is ultimately periodic. 48

Now: $\mathbb{Z}_2^L$ is finite. (In fact, $|\mathbb{Z}_2^L| = 2^L$)

$$\leadsto \exists j_1 < j_2. \sigma_{j_1} = \sigma_{j_2}$$

This means that after j_1 steps, we reach a state $\tau := \sigma_{j_1}$

which satisfies $\text{adv}^{j_2-j_1}(\tau) = \sigma_{j_2} = \sigma_{j_1} = \tau$

(under $\text{adv}^{j_2-j_1}(\tau)$)
apply adv a total
of j_2-j_1 times to τ

$\leadsto$ We may choose $C = j_1$, $N = j_2 - j_1$ //

Revised goal: Given that we can't escape ultimate periodicity,

try to design feedback shift registers with

"large" periods.

Defn A linear feedback shift register (LFSR) is a 49

feedback shift register with feedback function of the form

$$f(s_{j-1}, \dots, s_{j-L}) = c_1 s_{j-1} + \dots + c_L s_{j-L}$$

for fixed (!) $c_1, \dots, c_L \in \mathbb{Z}_2$.

The connection polynomial of the above LFSR is

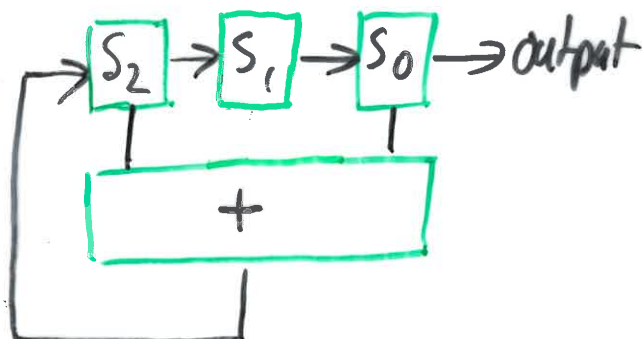
$$c(X) = 1 + c_1 X + \dots + c_L X^L,$$

a polynomial over $\mathbb{Z}_2$.

Note: LFSRs of a given length and their connection polynomials determine each other.

Ex: $L=3$, $c(X) = 1 + X + X^3$

Picture:



t	s_{t+2}	s_{t+1}	s_t	
0	1	0	1	initial state
1	0	1	0	
2	0	0	1	
3	1	0	0	
4	1	1	0	
5	1	1	1	
6	0	1	1	
7	1	0	1	initial state, again
8	0	1	0	

So the (least) period is $7 = 2^L - 1$.